

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2024
Introduction to Operating System (CS-222T)
Assignment # 1

Semester: 4th
Announced Date: 22-04-2024
Total Marks: 3
Instructor Name: Adeel Saeed

Batch: 2022F
Due Date: 29-04-2024
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 1	Explain fundamental operating system concepts such as processes, threads, CPU scheduling, deadlock.	PLO_2 (Knowledge for solving computing problem)	C2 (Understand)

Ques#1: Multi-core systems present certain challenges for multithreaded programming. Briefly describe these challenges.

Ques#2: What are the two separate modes of operation? Further, answer the followings:

- a. What is the mode of the system at boot time?
- b. What is the mode of the system when the operating system gains control?
- c. What is the mode of the system when a user program is running?

Ques#3: List five services provided by operating systems and explain how each service creates convenience for users. In which cases it would be impossible for user-level programs to provide these services? Explain your answer.